

Video Event Data Quality Proof of Concept

Objective

Develop a proof-of-concept solution to proactively identify data quality issues within video playback event data. The goal was to automate the detection of anomalies that could negatively impact downstream aggregations, reporting dashboards, and anomaly detection systems.

Approach

The solution was implemented using Python in Google Colab with pandas for data processing and analysis.

The development process involved:

Understanding the provided data dictionary and business constraints.

Translating those requirements into a set of automated validation rules.

Implementing checks to identify missing values, duplicate records, invalid field values, formatting inconsistencies, and business rule violations.

Generating both row-level findings and summary reports to make the results actionable for technical and non-technical stakeholders.

Throughout the implementation, emphasis was placed on keeping the code modular, readable, and easy to extend as new quality rules emerge.

How I Used AI in My Process

I used large language models (LLMs) as a development assistant throughout the exercise. AI was primarily used to clarify requirements, suggest additional edge cases to consider, improve the readability and organization of the code, and refine explanations within the reporting outputs. The overall solution design, validation logic, and final implementation decisions were made independently. All AI-assisted suggestions were reviewed, adapted where appropriate, and tested before being incorporated into the final deliverable.

How I Validated My Solution Works

The solution was validated by reviewing the generated row-level findings to ensure that the reported issues aligned with the defined business rules. Summary reports were compared against the detailed findings to verify consistency in issue counts. Additionally, manual spot checks were performed on selected records to confirm that the implemented validations correctly identified data quality issues. This process provided confidence in both the accuracy and completeness of the solution.

AI Best Practices Applied

Several best practices were followed while incorporating AI into the development process. AI-generated outputs were treated as suggestions rather than final answers, with all logic undergoing human review and verification. Clear and specific prompts were used to improve the quality of recommendations, while the final implementation prioritized simplicity, maintainability, and transparency. By combining AI-assisted productivity with engineering judgment and manual validation, the resulting solution remained both reliable and understandable.